



Muhammed Basith

Bachelor of Technology
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SUMMARY

Recent B.Tech graduate in Artificial Intelligence and Data Science with hands-on experience building full-stack AI applications. Proficient in Python, React, FastAPI, OpenAI API, and RAG systems. Built production-ready projects including a Multimodal RAG Assistant and AI-powered computer vision system. Seeking entry-level roles in AI Engineering or Full-Stack Development.

EDUCATION

•Bachelor of Technology in Artificial Intelligence And Data Science

2022-26

Ilahia College of Engineering and Technology, Mulavoor

PERSONAL PROJECTS

•AI Multimodal RAG Assistant

2026

Full-stack AI-powered multimedia question-answering system.

- Developed a full-stack AI application using **React**, **FastAPI**, and the **OpenAI API** to enable semantic question-answering over PDFs, audio, and video files.
- Integrated **Whisper AI** for multimedia transcription and implemented **timestamp-aware retrieval** with clickable playback navigation for relevant audio/video sections.
- Built a semantic vector search pipeline using **FAISS** and implemented automatic summarization, **MongoDB** persistence, Docker containerization, and CI/CD using GitHub Actions.
- **Technologies:** React, FastAPI, OpenAI API, Whisper, FAISS, MongoDB, Docker, GitHub Actions, Python, JavaScript. [\[GitHub\]](#)

•Smart Traffic Control System (SCTS)

2025

A website based Smart Traffic Control System.

- Engineered an AI-powered traffic signal control system using **YOLOv8**, achieving **95%+ vehicle detection accuracy** from traffic images.
- Implemented a dynamic signal timing algorithm that reduced simulated congestion by up to **40%**.
- Developed a responsive **Flask** web dashboard to visualize real-time vehicle counts and signal states.
- **Technologies:** Python, YOLOv8, OpenCV, Flask, HTML, CSS, JavaScript. [\[GitHub\]](#)

•Green Harvest - A Farmer Assistant

2026

Flask-based Agricultural Decision Support System

- Developed an end-to-end agricultural support platform integrating multiple **Machine Learning models** for crop recommendation, fertilizer suggestion, yield prediction, and disease detection.
- Implemented classification and regression models such as **Random Forest**, **Regression**, and **CNN** to achieve high prediction accuracy across diverse agricultural datasets.
- Preprocessed large-scale datasets (soil, crop, and weather) to improve model accuracy and reduce prediction latency.
- **Technologies:** Python, Flask, scikit-learn, TensorFlow, Pandas, OpenCV, HTML, CSS, JavaScript. [\[GitHub\]](#)

TECHNICAL SKILLS AND INTERESTS

Languages: Python, Java, JavaScript, HTML, CSS

Libraries Tools: OpenAI API, Whisper, FAISS, LangChain, YOLOv8, OpenCV, Flask, FastAPI, Docker

Web Dev Tools: React, REST API, Bootstrap

Cloud/Databases: MongoDB

Version Control: Git, GitHub, GitHub Actions

Relevant Coursework: DSA, OOP, OS, CN

Areas of Interest: Full-Stack Development, AI, Machine Learning

COURSES AND CERTIFICATIONS

•Front-End Development using Angular

May - July 2025

Nest Digital

Offline

- Completed a front-end development training focused on building responsive web interfaces using Angular.
- Gained practical experience in frontend development, including API integration and component-based architecture.